Red grapefruit positively influences serum triglyceride level in patients suffering from coronary atherosclerosis: studies in vitro and in humans.

The contents of the bioactive compounds in red and blond grapefruits and their influence on humans suffering from hypertriglyceridemia were studied. It was found that red grapefruit has a higher content of bioactive compounds and a higher antioxidant potential than blond grapefruit, determined by oxygen radical scavenging capacity, 1,1-diphenyl-2-picrylhydrazyl, carotenoid bleaching, and Folin-Ciocalteu assays. Fifty-seven hyperlipidemic patients, ages 39-72 years, after coronary bypass surgery, recruited from the Institute's pool of volunteers, were randomly divided into three equal in number (19) groups: two experimental (red and blond groups) and one control group (CG). During 30 consecutive days of the investigation the diets of the patients of the red and blond dietary groups were daily supplemented with one equal in weight fresh red or blond grapefruit, respectively. Before and after this trial, serum lipid levels of all fractions and serum antioxidant activity were determined. It was found that serum lipid levels in patients of the red and blond groups versus the CG after treatment were decreased: (a) total cholesterol, 6.69 versus 7.92 mmol/L, 15.5%, and 7.32 versus 7.92 mmol/L, 7.6%, respectively; (b) low-density lipoprotein cholesterol, 5.01 versus 6.29 mmol/L, 20.3%, and 5.62 versus 6.29 mmol/L, 10.7%, respectively; (c) triglycerides, 1.69 versus 2.32 mmol/L, 17.2%, and 2.19 versus 2.32 mmol/L, 5.6%, respectively. No changes in the serum lipid levels in patients of the CG were found. In conclusion, fresh red grapefruit contains higher quantities of bioactive compounds and has significantly higher antioxidant potential than blond grapefruit. Diet supplemented with fresh red grapefruit positively influences serum lipid levels of all fractions, especially serum triglycerides and also serum antioxidant activity. The addition of fresh red grapefruit to generally accepted diets could be beneficial for hyperlipidemic, especially hypertriglyceridemic, patients suffering from coronary atherosclerosis.